UNIT 1

Q1. What is the primary role of the OSI reference model in networking?

Answer:

The OSI model is a model that explains how computers communicate with each other.

- It divides network communication into 7 layers.
- Each layer performs a different job.
- It makes network communication easy to understand and manage.

Diagram:

```
7. Application
6. Presentation
5. Session
4. Transport
3. Network
2. Data Link
1. Physical
```

> **Example:** It is like dividing a big job into 7 small jobs.

Q2. List the advantages and disadvantages of Mesh Topology.

Answer:

Mesh topology is a network where every computer is connected to other computers through separate links.

Advantages:

- It is very reliable.
- If one connection fails, other connections work.
- It provides better security.

Disadvantages:

- It is very expensive.
- It needs more cables.
- It is difficult to install.

> **Example:** Computers in an important office network.

Q3. List and briefly explain any three functions of the Network Layer in OSI.

Answer:

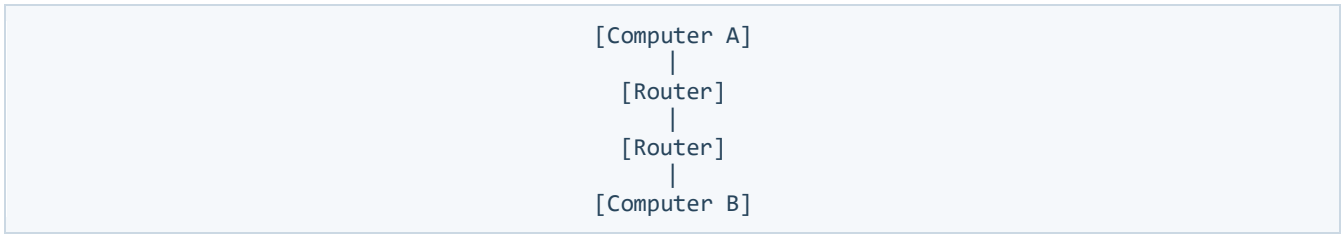
The Network Layer is the 3rd layer of the OSI model. It helps data travel from one network to another.

Functions:

- Logical Addressing: It gives an IP address to identify devices.

- Routing: It finds the best path to send data.
- Forwarding: It sends data to the next network device.

Diagram:



Example: Sending a message from one city to another using the internet.

Q4. State two key differences between the OSI Transport Layer and TCP/IP Transport Layer.

Answer:

Both transport layers help send data from one application to another.

OSI Transport Layer	TCP/IP Transport Layer
It is part of the OSI model.	It is part of the TCP/IP model.
It explains transport layer functions.	It uses TCP and UDP protocols.

- TCP provides reliable data delivery.
- UDP provides faster data delivery without guaranteeing delivery.

Example: TCP is used for reliable file transfer, while UDP is commonly used for online gaming.

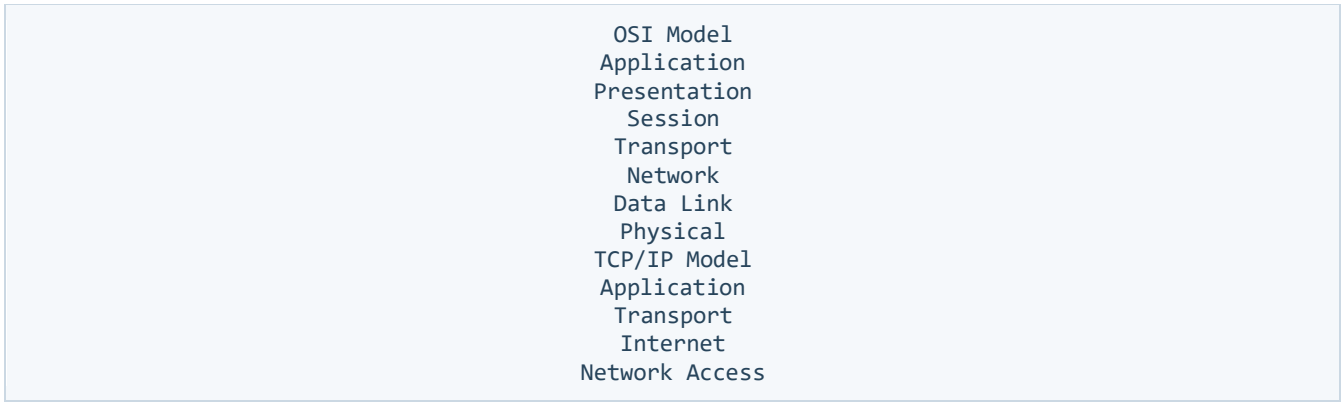
Q5. Mention two key differences between OSI and TCP/IP models.

Answer:

OSI and TCP/IP are models used to understand network communication.

OSI Model	TCP/IP Model
It has 7 layers.	It has 4 layers in the common model.
It is a reference model.	It is a practical communication model.

Diagram:



Example: Both models explain how data travels between computers.

Q6. What is coaxial cable and where is it commonly used?

Answer:

A coaxial cable is a type of wire used to send data and signals.

- It has a central copper wire.
- It has insulation and a protective outer cover.
- It is commonly used in cable TV and internet connections.

Example: Cable TV connection in our homes.

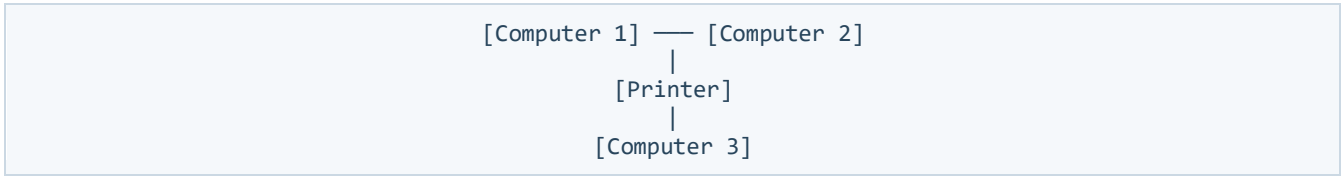
Q7. Define a Computer Network.

Answer:

A computer network is a group of two or more computers connected together to share information and resources.

- It helps to share files.
- It helps to share printers.
- It allows communication between computers.

Diagram:



Example: Computers connected in a college computer lab.

Q8. What are LAN, MAN and WAN?

Answer:

LAN, MAN and WAN are three types of computer networks.

- LAN (Local Area Network): Connects computers in a small area like a home or school.

- MAN (Metropolitan Area Network): Connects computers across a city.
- WAN (Wide Area Network): Connects computers across large areas like countries.

Q9. List the different types of Network Topologies.

Answer:

Network topology means the way computers are connected in a network.

The different types are:

- Bus Topology: All computers are connected to one main cable.
- Star Topology: All computers are connected to one central device.
- Ring Topology: Computers are connected in a circle.
- Mesh Topology: Computers are connected through many separate links.
- Tree Topology: Computers are connected like branches of a tree.
- Hybrid Topology: Combination of two or more topologies.

Q10. What is the difference between Guided and Unguided Transmission Media?

Answer:

Transmission media is the way of sending data from one device to another.

Guided Media	Unguided Media
Data travels through cables.	Data travels without cables.
It uses a physical path.	It uses air or space.
Example: Optical fibre, twisted pair cable.	Example: Wi-Fi, radio waves.

Unit 2

Q1. Enumerate any four responsibilities of the Data Link Layer in OSI Model.

Answer:

The Data Link Layer is the 2nd layer of the OSI model. It helps to transfer data safely between two connected devices.

Four responsibilities:

- Framing: It divides data into small parts called frames.
- Error Control: It finds errors in the received data.
- Flow Control: It controls the speed of sending data.
- Access Control: It decides which device can send data.

Diagram:

```
Data Link Layer
    Framing
 Error Control
 Flow Control
Access Control
```

> **Example:** It helps two computers send data correctly.

Q2. What are the different types of Errors that occur in the data in a network? Give Example.

Answer:

An error occurs when the received data is different from the sent data.

There are two main types of errors:

1. Single Bit Error: Only one bit in the data changes from 0 to 1 or 1 to 0.

Example:

- Sent: `1010`
- Received: `1000`

2. Burst Error: Two or more bits in the data are changed.

Example:

- Sent: `101010`
- Received: `110110`

Diagram:

```
1. Single Bit Error
      Sent
      1010
   Received
      1000
2. Burst Error
      Sent
     101010
   Received
     110110
```

> **Example:** A wrong bit during data transfer is called an error.

Q3. Explain the importance of framing in the Data Link Layer.

Answer:

Framing is the process of dividing large data into small parts called frames.

Importance of Framing:

- It divides data into small frames.
- It helps the receiver identify the beginning and ending of data.
- It makes data transfer easy and organized.

Diagram:

```
Original Data
  Large Data
After Framing
   Frame 1
   Frame 2
   Frame 3
```

> **Example:** Sending a big message by dividing it into small parts.

Q4. Define Flow Control and explain why it is essential?

Answer:

Flow control is a method of controlling the speed of data transmission between two devices.

Why is it essential?

- It controls the speed of sending data.
- It prevents the receiver from getting too much data.
- It helps the receiver receive data properly without losing it due to overload.

Diagram:

```
[Sender] ——— Data ———> [Receiver]
     |                      |
     | <——— Control ——————— |
```

> **Example:** If a sender sends data very fast, the receiver may not handle it. Flow control manages the speed.

Q5. What is One's Complement Checksum? Where is it used?

Answer:

One's Complement Checksum is a method used to find errors in data during transmission.

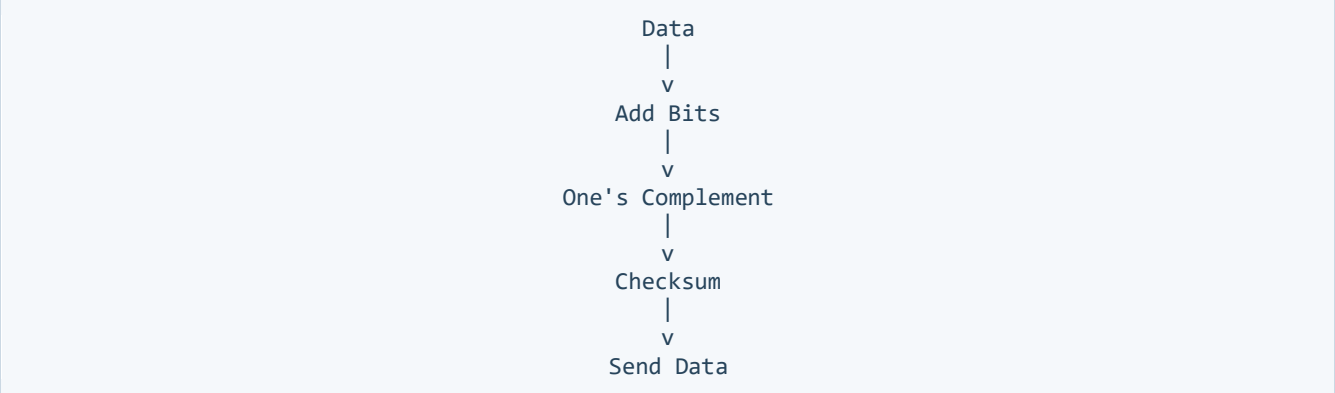
Working:

- First, the sender adds all data bits.
- Then it changes 0 to 1 and 1 to 0. This is called one's complement.
- The checksum is sent along with the data to check for errors.

Where is it used?

It is used in IP, TCP and UDP protocols.

Diagram:



> **Example:** It works like checking a total amount to find mistakes.

Q6. What is CRC (Cyclic Redundancy Check)?

Answer:

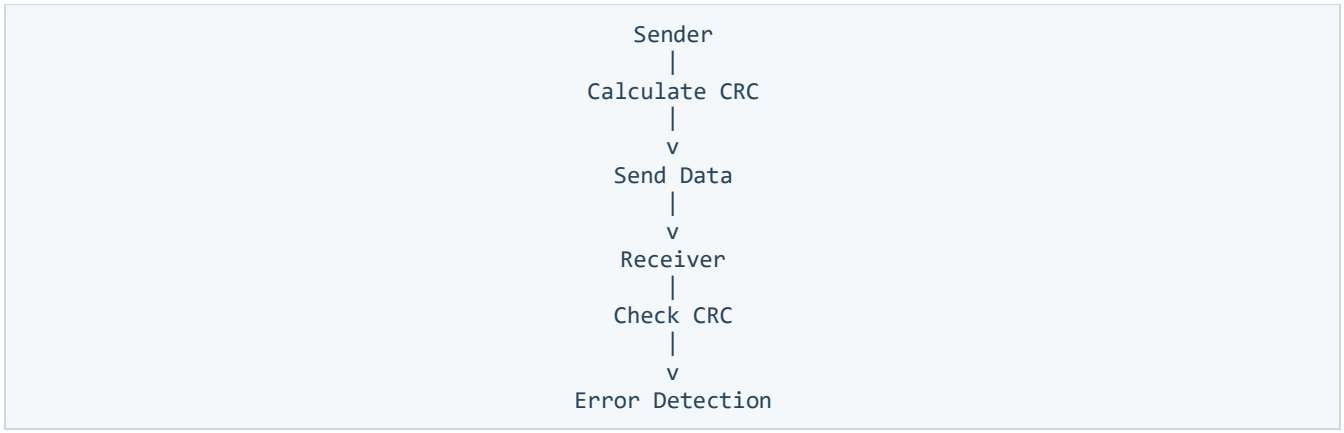
CRC stands for Cyclic Redundancy Check.

It is a method used to find errors in data during transmission.

Working:

- The sender calculates a special value called CRC.
- It adds this value to the data and sends it.
- The receiver checks the CRC value to find errors.

Diagram:



> **Example:** CRC works like checking whether the received data is correct or not.

Q7. Explain the basic idea of the Stop-and-Wait protocol.

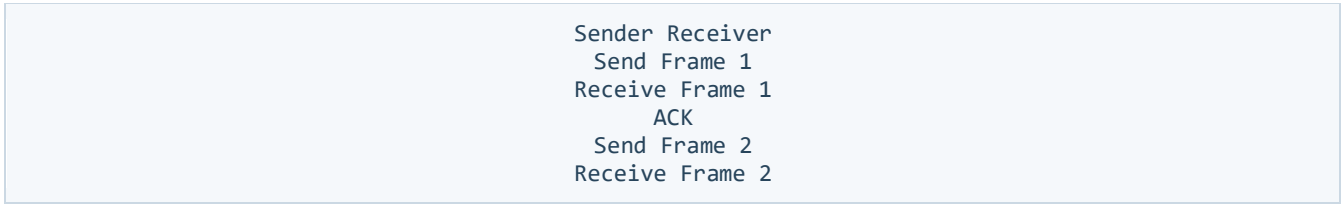
Answer:

Stop-and-Wait is a method used to send data one frame at a time.

Working:

- The sender sends one frame to the receiver.
- The sender waits for an acknowledgement (ACK).
- After receiving ACK, the sender sends the next frame.

Diagram:



> **Example:** It is like sending one message and waiting for a reply before sending the next message.

Unit -3

Q1. Briefly explain the core difference between Random Access and Controlled Access protocols.

Answer:

Random Access and Controlled Access are two methods used to send data in a network.

- Random Access: Any device can send data whenever it wants. Two devices may send data at the same time.
- Controlled Access: Devices send data one by one with permission.
- Random Access may cause collisions, but Controlled Access reduces collisions.

Example:

- Random Access: Everyone speaking at the same time.
- Controlled Access: Everyone speaking one by one.

Q2. Describe the working of Pure ALOHA protocol.

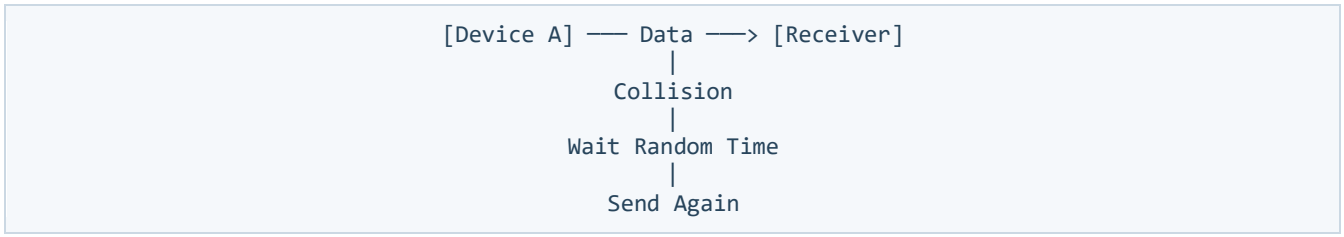
Answer:

Pure ALOHA is a Random Access protocol used to send data in a network.

Working:

- A device sends data whenever it is ready.
- If two devices send data at the same time, a collision occurs.
- If collision occurs, the device waits for a random time and sends the data again.

Diagram:



> **Example:** Two students speaking at the same time and then trying again one by one.

Q3. What is CSMA and how does it reduce collisions?

Answer:

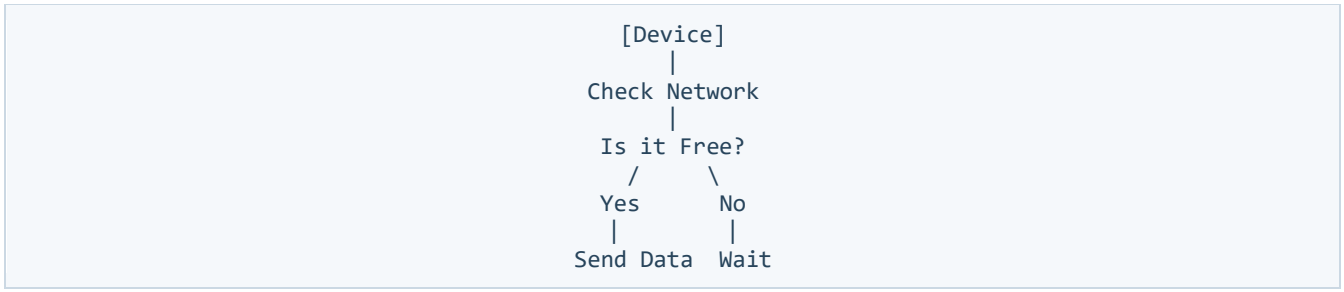
CSMA stands for Carrier Sense Multiple Access.

It is a method used to reduce collisions while sending data.

Working:

- First, the device checks whether the network is free or busy.
- If the network is free, it sends data.
- If the network is busy, it waits until the network becomes free.

Diagram:



Example: Before speaking, we listen to check whether someone else is speaking.

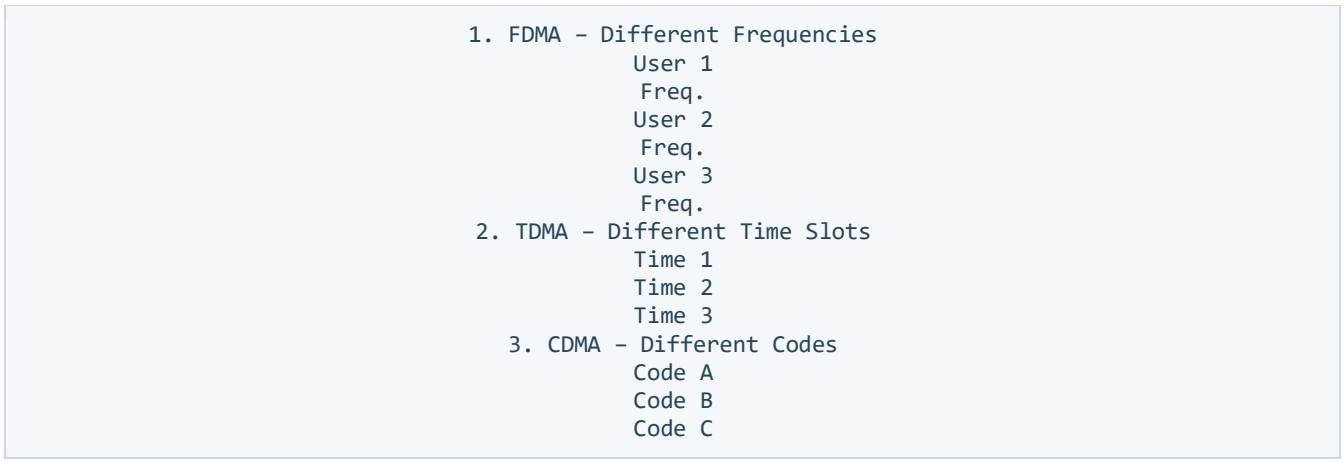
Q4. Compare FDMA, TDMA, and CDMA in Channelization.

Answer:

FDMA, TDMA and CDMA are three channelization techniques used to share a communication channel.

FDMA	TDMA	CDMA
Divides channel by frequency.	Divides channel by time.	Uses different codes.
Each user gets a separate frequency.	Each user gets a separate time slot.	Each user gets a unique code.

Simple Diagram:



Example: Mobile phones use these techniques to share communication resources.

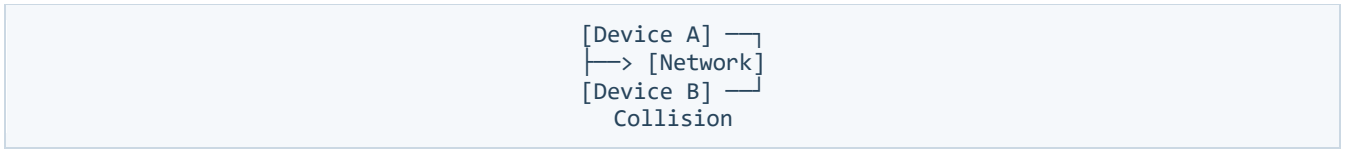
Q5. What is Random Access in MAC protocols?

Answer:

Random Access is a method in which any device can send data whenever it wants.

- Every device can try to send data.
- Two or more devices may send data at the same time.
- If a collision occurs, the devices wait and try again.

Diagram:



Examples: Pure ALOHA, Slotted ALOHA and CSMA.

Q6. Define Carrier Sense Multiple Access (CSMA).

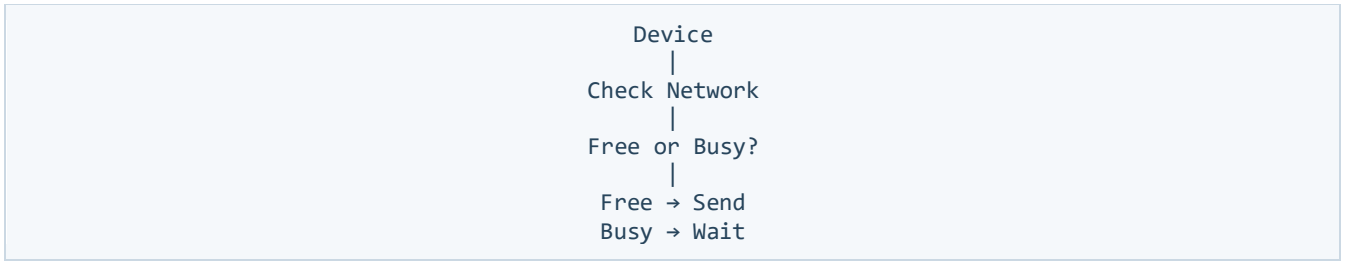
Answer:

CSMA stands for Carrier Sense Multiple Access.

It is a network method used to reduce data collisions.

- First, a device listens to the network.
- If the network is free, it sends data.
- If the network is busy, it waits.

Diagram:



Example: Like waiting for another person to finish speaking before we speak.

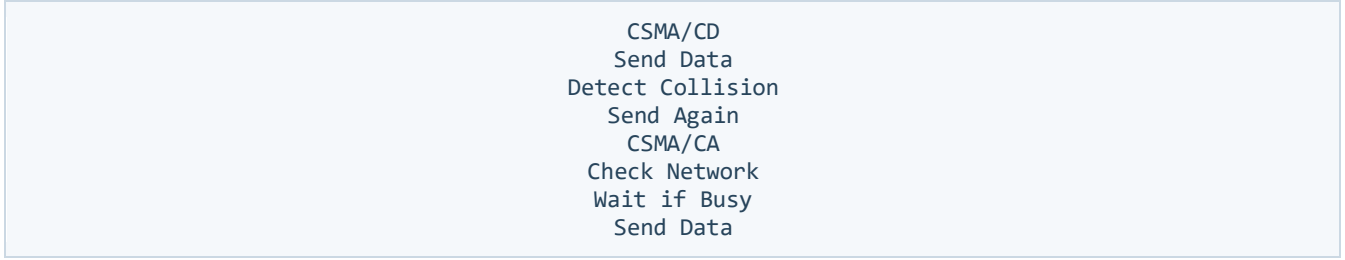
Q7. Differentiate between CSMA/CD and CSMA/CA.

Answer:

CSMA/CD and CSMA/CA are two methods used to handle data collisions in networks.

CSMA/CD	CSMA/CA
CD means Collision Detection.	CA means Collision Avoidance.
It detects collisions after they happen.	It tries to avoid collisions before they happen.
Used in traditional wired Ethernet.	Commonly used in Wi-Fi.

Diagram:



Example:

- CSMA/CD: Finds a collision after it happens.
- CSMA/CA: Tries to prevent a collision before sending data.

Q8. What is Channelization? Name the channelization techniques.

Answer:

Channelization is a method of dividing one communication channel among multiple users.

It allows many users to share the same communication channel.

Three channelization techniques:

- FDMA: Divides the channel based on frequency.
- TDMA: Divides the channel based on time.
- CDMA: Divides users based on different codes.

Diagram:

```
Channelization
     FDMA
     TDMA
     CDMA
```

> **Example:** Many mobile users sharing communication resources.

Q9. List the services provided by the Network Layer to the Transport Layer.

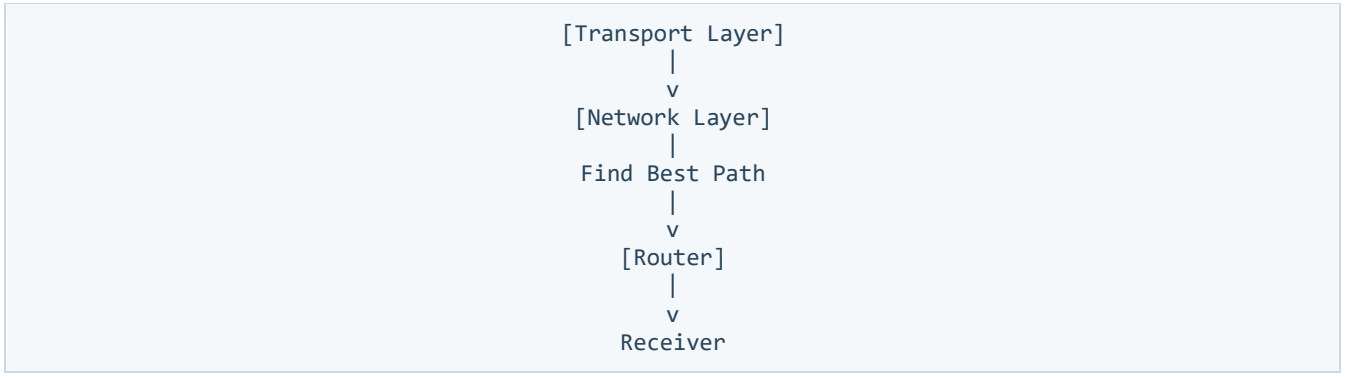
Answer:

The Network Layer is the 3rd layer of the OSI model. It helps transfer data from one network to another.

Services:

- Logical Addressing: Identifies devices using IP addresses.
- Routing: Finds the best path to send data.
- Packet Delivery: Sends packets from the sender to the receiver.
- Fragmentation: Divides large packets into smaller parts when needed.

Diagram:



> **Example:** When we open a website, the Network Layer helps packets travel from our device to the website server.

Unit-4

Q1. Explain the working of the Shortest Path Routing Algorithm.

Answer:

Shortest Path Routing is a method used to find the shortest and best path to send data from one computer to another.

Working:

- First, it finds all possible paths between the sender and receiver.
- It calculates the cost of each path.
- It selects the path with the lowest cost.
- Data is sent through that path.

Diagram:

> **Example:** Google Maps finds the shortest route to reach a destination.

Q2. What is the Count-to-Infinity problem in Distance Vector Routing?

Answer:

Count-to-Infinity is a problem that occurs in Distance Vector Routing.

- When a network connection fails, routers may get wrong information.
- Routers keep increasing the distance to the failed network.
- This continues for a long time until the problem is solved.

> **Example:** Two friends keep telling each other that a lost person can be found through the other friend.

> **Remember:** Count-to-Infinity means routers keep increasing the distance to a failed network.

Q3. What is Hierarchical Routing and where is it used?

Answer:

Hierarchical Routing is a method of dividing a large network into small groups called regions or areas.

- It divides a large network into smaller areas.
- Each area manages its own network.
- It makes routing easy and reduces the work of routers.

Diagram:

```
Large Network
   Area 1
   Area 2
   Area 3
```

Where is it used?

It is used in large networks like the Internet.

> **Example:** Dividing a big school into different departments.

Q4. What are the general principles of Congestion Control?

Answer:

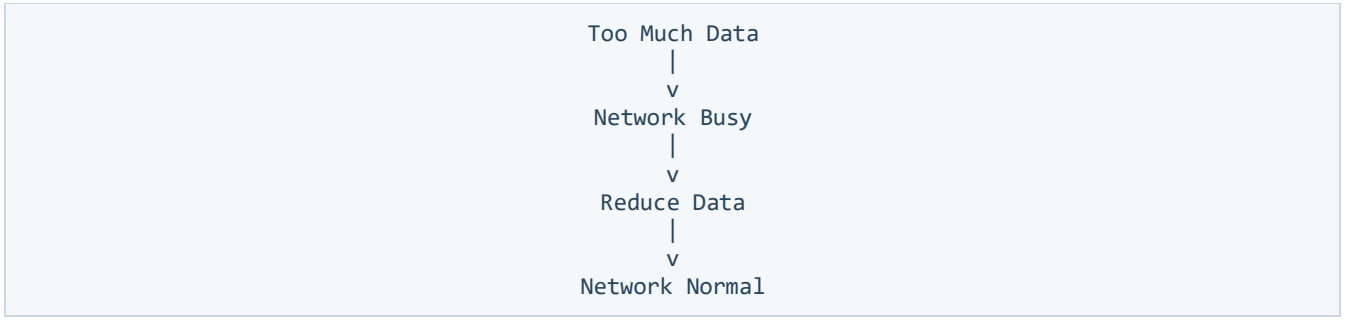
Congestion means too much data is present in a network at the same time.

Congestion Control is a method used to reduce heavy traffic in a network.

General Principles:

- Prevent Congestion: Avoid sending too much data into the network.
- Detect Congestion: Find when the network becomes busy.
- Control Congestion: Reduce data traffic to make the network work properly.

Diagram:



> **Example:** If too many vehicles are on a road, traffic police control the traffic.

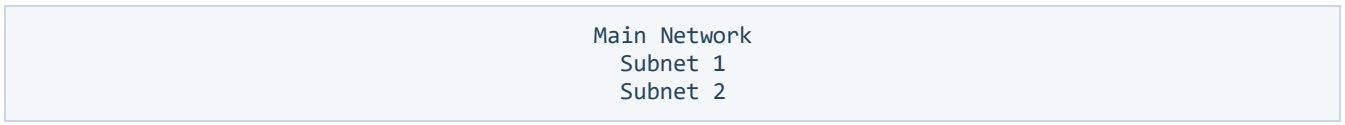
Q5. Define a Subnet.

Answer:

A subnet is a small network created by dividing a large network into smaller networks.

- It divides a large network into small parts.
- Each subnet has its own network address.
- It makes network management easy and organized.

Diagram:



> **Example:** A college network divided into separate networks for the library and computer labs.

Q6. What is the difference between Open-Loop and Closed-Loop Congestion Control?

Answer:

Both are methods used to control network congestion.

Open-Loop Congestion Control	Closed-Loop Congestion Control
It prevents congestion before it happens.	It controls congestion after detecting it.
It does not use feedback.	It uses feedback from the network.
It follows fixed rules.	It changes its actions based on network conditions.

Example:

- Open-Loop: Taking steps to avoid traffic before it happens.
- Closed-Loop: Controlling traffic after seeing a traffic jam.

Q7. What is Tunneling in Computer Networks?

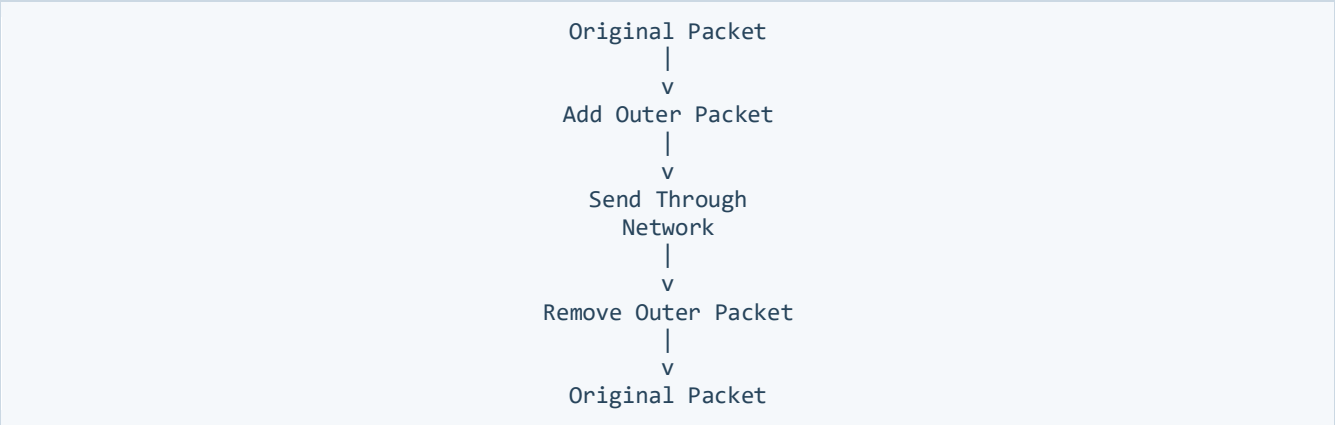
Answer:

Tunneling is a method of sending one type of network data inside another type of network packet.

Working:

- The sender takes the original data packet.
- It puts that packet inside another packet. This is called encapsulation.
- The packet travels through the network and reaches the receiver.

Diagram:



> **Example:** VPN uses tunneling to send data securely through the Internet.

UNIT-5

Q1. Differentiate between Connection-Oriented and Connectionless Transport Protocols.

Answer:

These are two methods used to send data from one device to another.

Connection-Oriented	Connectionless
First, it creates a connection between devices.	It does not create a connection.
It provides reliable data delivery.	It does not guarantee data delivery.
It is slower compared to connectionless communication.	It is generally faster.
Example: TCP	Example: UDP

Example:

- TCP: Like making a phone call before talking.
- UDP: Like sending a letter without waiting for a reply.

Q2. Explain the Structure of a UDP Segment.

Answer:

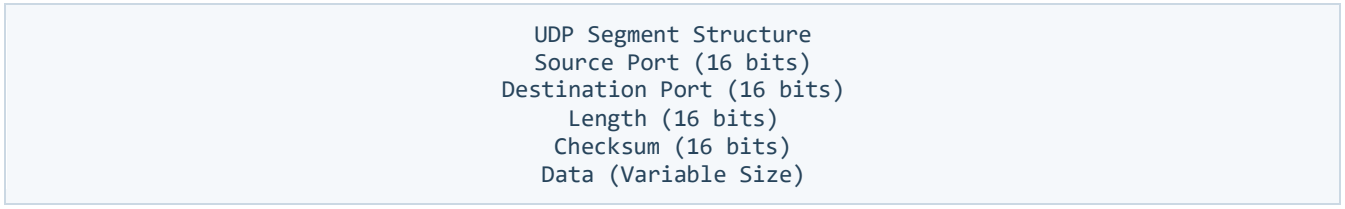
UDP stands for User Datagram Protocol.

A UDP segment contains a header and data. The header has four parts.

Four parts of UDP Header:

- Source Port: Identifies the sender's application.
- Destination Port: Identifies the receiver's application.
- Length: Tells the total size of the UDP segment.
- Checksum: Helps to find errors in data.

Diagram:



> **Example:** UDP is used in online games and video calls.

Q3. What are the Key Features of TCP?

Answer:

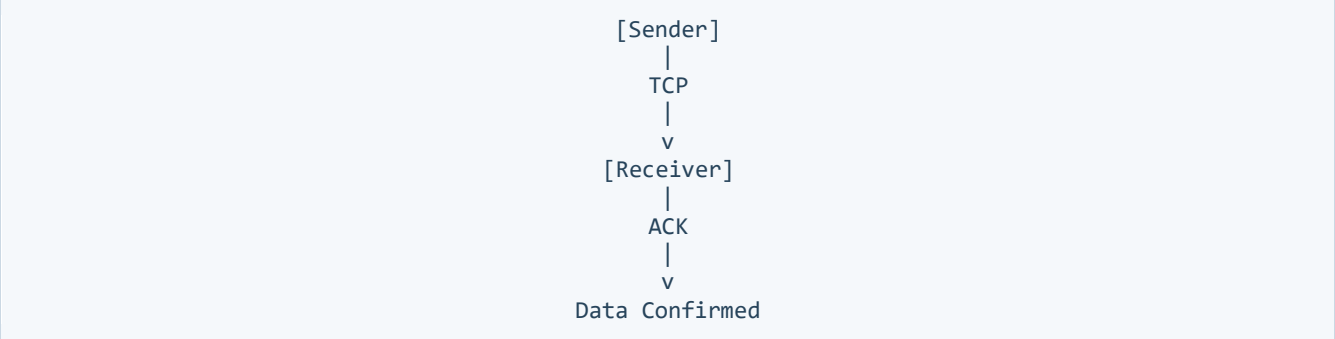
TCP stands for Transmission Control Protocol.

It is used to send data safely from one device to another.

Key Features:

- Reliable Delivery: It makes sure data reaches the receiver.
- Error Control: It finds and corrects transmission errors by retransmitting lost data.
- Flow Control: It controls the speed of sending data.
- Ordered Delivery: It delivers data in the correct order.

Diagram:



> **Example:** TCP is used when downloading files and opening websites using HTTPS.

Q4. How does TCP achieve Error Control?

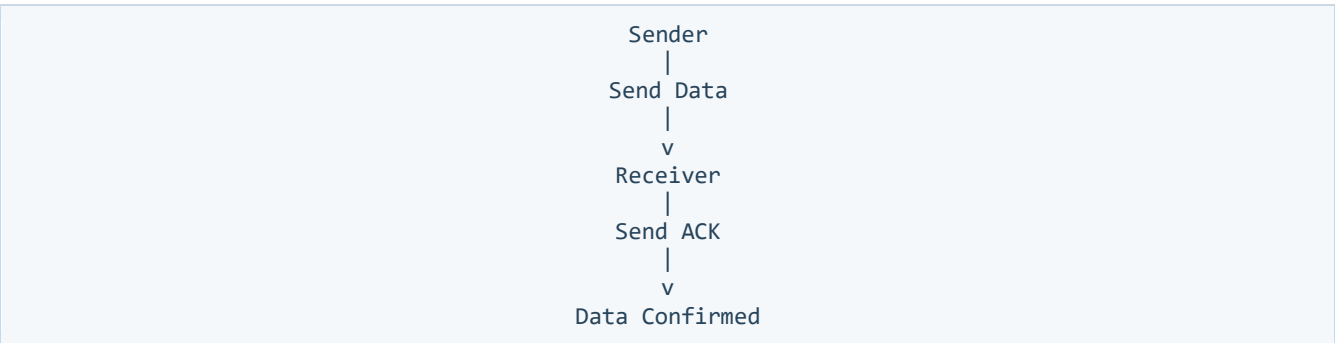
Answer:

TCP Error Control is a method used to find and correct errors during data transmission.

Working:

- TCP uses a Checksum to find errors in data.
- The receiver sends an ACK (Acknowledgement) after receiving data.
- If data is lost or damaged, TCP sends it again. This is called Retransmission.

Diagram:



> **Example:** If a file is not received correctly, TCP sends the missing data again.

Q5. What is the Role of the Transport Layer in the OSI Model?

Answer:

The Transport Layer is the 4th layer of the OSI model. It helps transfer data between applications on different devices.

Main Roles:

- Segmentation: Divides large data into small parts.
- Error Control: Finds errors and helps deliver correct data.
- Flow Control: Controls the speed of data transfer.
- Process Delivery: Sends data to the correct application.

Diagram:

```
Sender Application
Transport Layer
Receiver Application
```

> **Example:** It helps send a message from one mobile application to another.

Q6. Differentiate between Local and Remote Login.

Answer:

Local Login and Remote Login are two ways of accessing a computer.

Local Login	Remote Login
We access a computer directly.	We access another computer through a network.
We use the same computer.	We use another computer from a different place.
No network is needed.	A network connection is needed.
Example: Logging into our own laptop.	Example: Accessing an office computer from home.

> **Remember:** Local = Same computer. Remote = Another computer through a network.

Q7. Name Two Services Provided by the UDP Protocol.

Answer:

UDP stands for User Datagram Protocol.

It provides the following services:

- Process-to-Process Delivery: It sends data from one application to another application.
- Error Detection: It uses a checksum to check whether data has errors.

> **Other important point:** UDP is fast because it does not wait for acknowledgements.

> **Example:** UDP is used in video calls, online games and live streaming.

Q8. Define TELNET in the Context of the Application Layer.

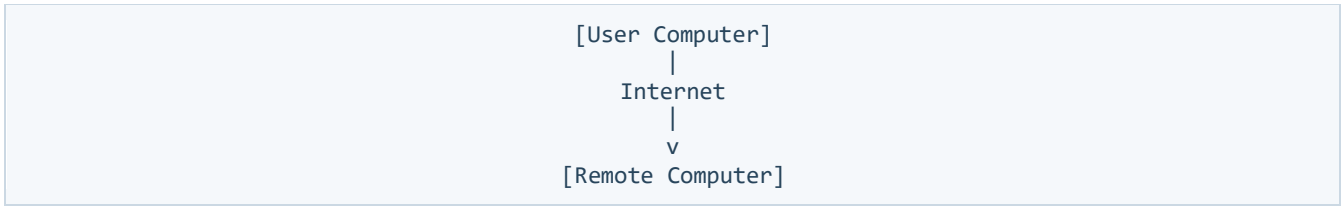
Answer:

TELNET stands for TELecommunication NETwork.

It is an Application Layer protocol used to access another computer remotely.

- It allows users to control another computer from a different place.
- It works using a network connection.
- It uses port number 23.

Diagram:



Example: A person can access and control an office computer from home using TELNET.

Note: TELNET is not secure because it sends data, including passwords, without encryption.